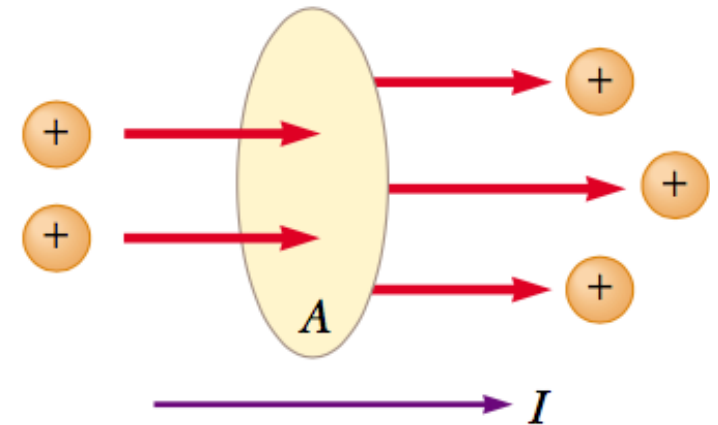


27.1 Electric Current

Charges in motion through an area A . The time rate at which charge flows through the area is defined as the current I . The direction of the current is the direction in which positive charges flow when free to do so.

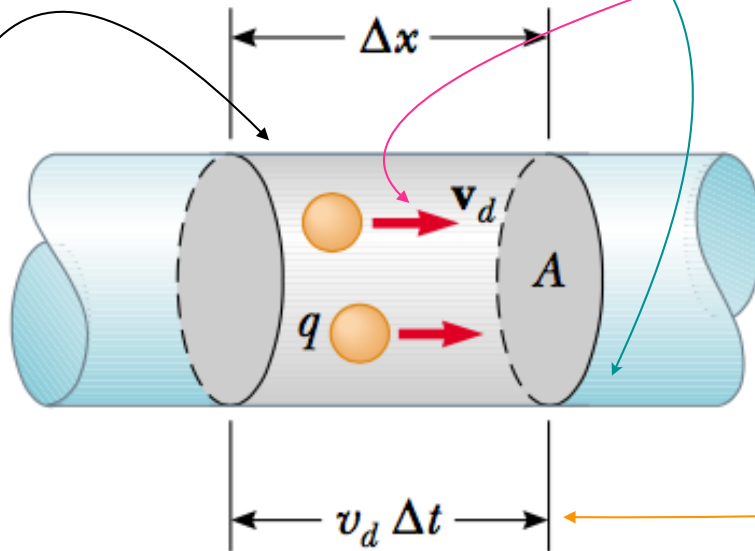


The current is the rate at which charge flows through this surface. $I_{\text{av}} = \frac{\Delta Q}{\Delta t}$

instantaneous current $I \equiv \frac{dQ}{dt}$

Microscopic Model of Current

A section of a uniform conductor of cross-sectional area A



The mobile charge carriers move with a speed v_d

the displacement they experience in the x direction in a time interval Δt is $\Delta x = v_d \Delta t$

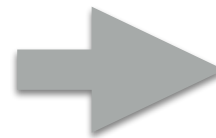
The volume of the grey section is $A \cdot \Delta x$

If n represents the number of mobile charge carriers per unit volume (in other words, the charge carrier density) the number of carriers in the gray section is $nA\Delta x$.

Therefore, the total charge ΔQ in this section is

$$\Delta Q = \text{number of carriers in section} \times \text{charge per carrier} = (nA \Delta x) q$$

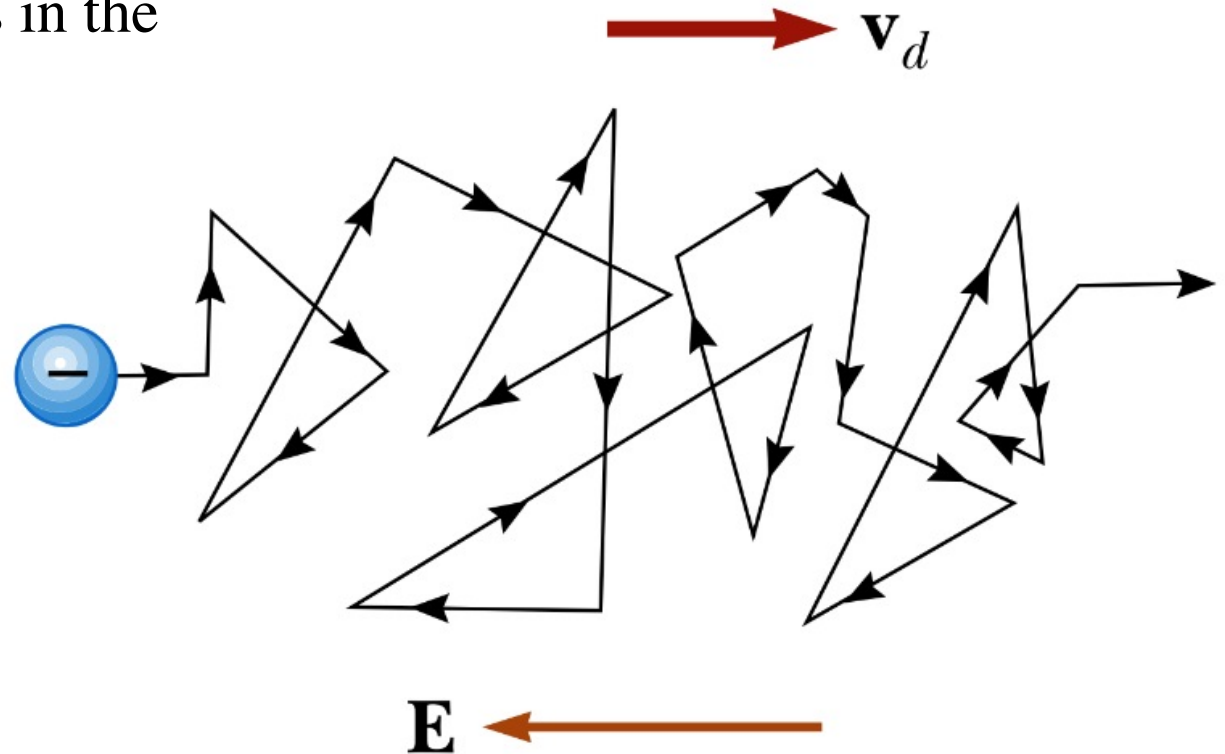
$$\Delta Q = (nA v_d \Delta t) q$$



$$I_{\text{av}} = \frac{\Delta Q}{\Delta t} = nq v_d A$$

A schematic representation of the zigzag motion of an electron in a conductor. The changes in direction are the result of collisions between the electron and atoms in the conductor.

Note that the net motion of the **electron is opposite** the direction of the **electric field**.



Because of the acceleration of the charge carriers due to the electric force, the paths are actually parabolic. However, the **drift speed** is much smaller than the average speed, so the parabolic shape is not visible on this scale.

27.2 Resistance

The current density J

$$J \equiv \frac{I}{A} = nqv_d$$

replacing

$$I_{av} = \frac{\Delta Q}{\Delta t} = nqv_d A$$

Current density

$$\mathbf{J} = nq\mathbf{v}_d$$

A current density $\mathbf{J}$ and an electric field $\mathbf{E}$ are established in a conductor whenever a potential difference is maintained across the conductor.

$$\mathbf{J} = \sigma \mathbf{E}$$

the constant σ of proportionality is called the **conductivity** of the conductor

Ohm's law

for many materials (including most metals), the ratio of the current density to the electric field is a constant σ that is independent of the electric field producing the current.



Georg Simon Ohm

German physicist (1789–1854)

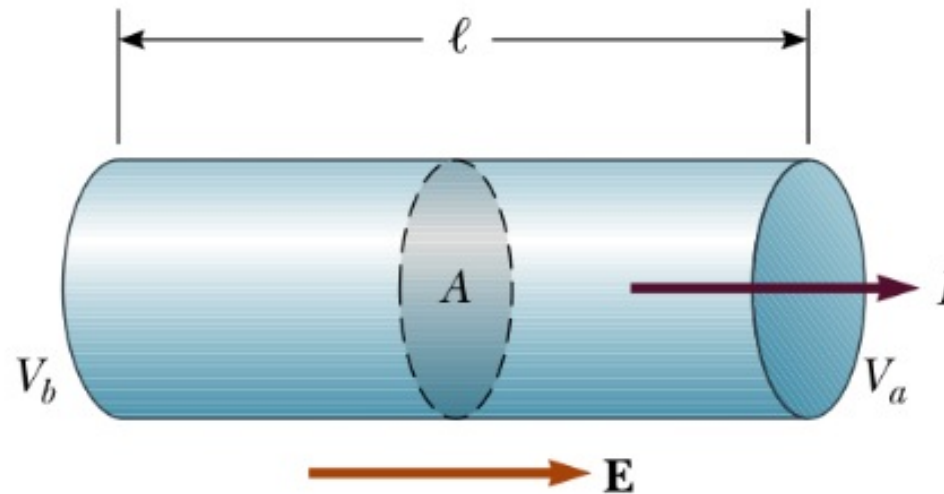


Figure 27.5 A uniform conductor of length ℓ and cross-sectional area A . A potential difference $\Delta V = V_b - V_a$ maintained across the conductor sets up an electric field $\mathbf{E}$, and this field produces a current I that is proportional to the potential difference.

$$\Delta V = V_b - V_a$$

Therefore, we can express the magnitude of the current density in the wire as

$$J = \sigma E = \sigma \frac{\Delta V}{\ell}$$

$$\Delta V = E\ell$$

Because $J = I/A$,

$$\Delta V = \frac{\ell}{\sigma} J = \left(\frac{\ell}{\sigma A} \right) I = RI$$

The quantity $R = \ell / \sigma A$ is called the **resistance** of the conductor. '

$$R \equiv \frac{\Delta V}{I} \quad 1 \, \Omega \equiv \frac{1 \, \text{V}}{1 \, \text{A}}$$

The inverse of conductivity is **resistivity**³ ρ :

$$\rho = \frac{1}{\sigma}$$

$$R = \rho \frac{\ell}{A}$$

Resistance of a uniform material along the length

27.4 Resistance and Temperature

$$\rho = \rho_0[1 + \alpha(T - T_0)]$$

Variation of with temperature

$$\alpha = \frac{1}{\rho_0} \frac{\Delta \rho}{\Delta T}$$

Temperature coefficient of resistivity

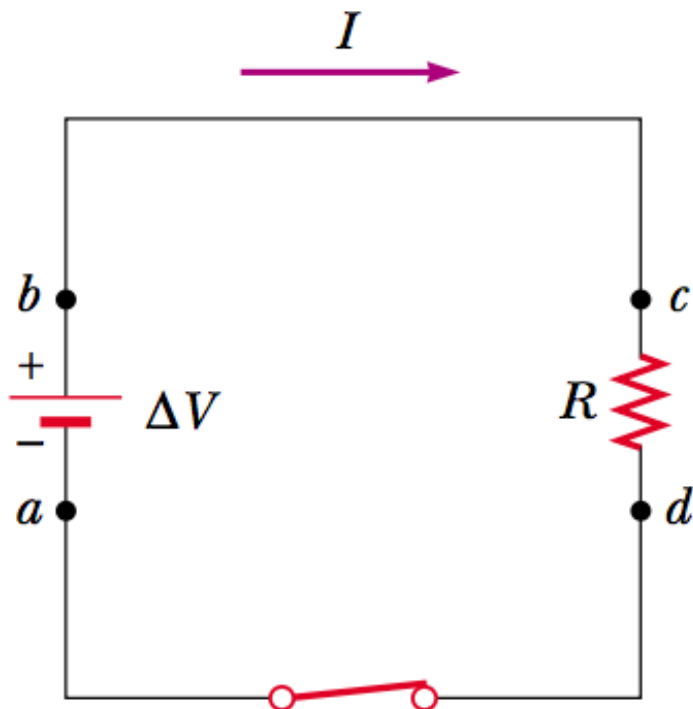
$$R = R_0[1 + \alpha(T - T_0)]$$

The resistance of the material increases with the temperature
higher the T, higher the R

27.6 Electrical Power

A circuit consisting of a resistor of resistance R and a battery having a potential difference ΔV across its terminals.

Positive charge flows in the clockwise direction.



Let us consider now the rate at which the system loses electric potential energy as the charge Q passes through the resistor:

$$\frac{dU}{dt} = \frac{d}{dt} (Q\Delta V) = \frac{dQ}{dt} \Delta V = I\Delta V$$

Power delivered to a device

$$\mathcal{P} = I\Delta V$$

Power delivered to a device

$$\mathcal{P} = I^2 R = \frac{(\Delta V)^2}{R}$$